infrastructure / audit · claim holds · holds, qualified · withdrawn / failed · provisional / open

PHASE 1 · build, then freeze

the whole SH3 family held out — not random variants, because two variants in one domain are not independent

MegaScale
389,068 substitutions
412 domains, 32–72 aa
ΔΔG by proteolysis

Features · 1,292
1,280 ESM-2 650M dims
+ 12 scalars
structure excluded by design

Head · 1292→512→128→1
frozen encoder, small head
grouped on accession
SH3 family held out

Conformal calibration
q90 = 6.807
coverage 89.9 % vs 90 %
fold_lo ≥ 1 = confident

Held-out accuracy
median ρ = 0.839
15 SH3 domains
beats zero-shot ESM-2 15/15

Applied to the atlas
6,884 cancer variants
310 domains, 235 genes
410 confident calls (6.0 %)

FROZEN · input/frozen/ is chmod a-w · everything below re-derives and asserts first · 187/187 values reproduce

PHASE 2 · test it against the outside world

four independent external sources — and a re-test of the project's own claims

X0 · acquire once
ClinVar, gnomAD, Domainome
assert row count unchanged
→ caught 3 join defects

X1 · dbNSFP
2.1 M rows → 152,852
coverage vs publisher's own
→ caught mislabelled columns

Pre-registration
rule committed to git
BEFORE the code existed
threshold, min n, reading

E2 · measured
Domainome, 880 variants
41 domains, 32 genes
|ρ| 0.600 = 85 % of the
0.709 two-assay ceiling

E3 · benchmark
ThermoMPNN disqualified:
13/15 domains in its training
SaProtΔG 0.855 vs 0.839 (ns)
ESM3ΔG 0.792, p = 0.0084

E1 · clinical
ClinVar 21 vs 101
1.67 vs 0.19 kcal/mol
$p = 4.9 \times 10^{-6}$
but sensitivity only 4.8 %

E4 · specificity
model r = +0.023 (p 0.91)
CONTROL: observed profile
r = −0.051 — a predictor
of X cannot beat X

G1 · context-aware null
6,456 coords, 100 % verified
0.0923 → 0.0860 → 0.0788
→ 0.0604 vs observed 0.0596
p = 0.40, two spectra agree

G2 · per-domain intervals
coverage 83.3 % – 95.0 %
quantile varies 1.79×
count robust: 410 (378–421)
per-variant claim is not

G3 · scoped, then dropped
would recover 72, not 219
changes NO prediction
→ kept as a limitation

G4 · rebuild the panel
5/10 picks at one position
controls differed by burial
→ 10 positions, 3 exposed

OUTCOME · three objections closed · one claim WITHDRAWN — the cancer depletion was mutational bias, not selection

PHASE 3 · does any of it generalise?

provisional — not in the manuscript

H1.0 · leakage audit
every other family WAS
in training; PDZ absent only
because it exceeds 72 aa

H1.1 · power first
15 domains → CI width 0.243
wider than the margin itself
→ pooled to 35, width 0.148

H1.2 · evaluation
PDZ 0.602 vs SH3 0.623
three families 0.420
main arm INCONCLUSIVE

H1.3 · assay quality
matched: −0.203 → −0.187
Myb/SANT: better data,
lower accuracy

H2 · bench
20 constructs ready (G4)
DSF, then CD on discordant
NOT STARTED

Length is NOT the limit — PDZ is 88 aa against a 72 aa training ceiling and still matches SH3.
Measurement quality is not the limit either. What is, is not yet named.